

1/13

TechPulse

Tech Job Market Intelligence Platform

Real-time insights about tech jobs, skills, companies and locations.

Built with passion for data, technology and innovation.

2/13



Project Overview

TechPulse is a data-driven platform that collects, analyzes and visualizes technology job market data to provide real-time insights and help people understand the tech job market better.

We solve the problem of scattered and unstructured job data by bringing it together in one intelligent platform.



Real-time Job Data
Collecting jobs from multiple sources



Actionable Insights
Turning data into useful intelligence



Smarter Decisions
Helping job seekers and companies

3/13



Data Collection

We collect job data from multiple sources using Python.



Python Scrapers
Custom web scrapers



APIs
REST APIs integration



Scheduled Jobs
Automated collection

Tools & Libraries

BeautifulSoup • Selenium
Requests • Pandas

4/13



Data Cleaning & Processing

Raw data goes through multiple cleaning and transformation steps.



Data Cleaning
Removing nulls and errors



Data Validation
Ensuring data accuracy



Deduplication
Removing duplicate records



Standardization
Standardizing formats and values

Tools: Python • Pandas

5/13



SQL Server & Data Warehouse

Cleaned data is stored in SQL Server and structured into a dimensional data warehouse.



SQL Server

- Raw Data
- Cleaned Data
- Staging Area

Dimensional Model

- FactJobs
- FactJobSkills
- DimCompany
- DimLocation
- DimPlatform
- DimSkills
- DimDate

6/13



SSIS ETL Pipeline

We use SQL Server Integration Services (SSIS) to load data into the data warehouse.



Extract
Get data from source



Transform
Clean and prepare data



Lookup
Check existing records



Incremental Load
Insert only new records



Prevent Duplicates
Ensure data quality



SSIS
Incremental Load with Lookup

Efficient • Reliable • Scalable

7/13



Power BI Dashboard

Interactive dashboards that turn data into powerful insights.



Total Jobs
2,965



Companies
7



Platforms
3



Locations
540

Dashboards Include

- Jobs Overview
- Experience Level
- Top Companies
- Work Type
- Jobs Trend Over Time
- Data Quality
- Skills Analysis
- And More...



Power BI • DAX • Interactive Reports

8/13



Backend API

A robust RESTful API built with Node.js and Express.js.



- ✓ JWT Authentication
- ✓ Rate Limiting
- ✓ Data Validation
- ✓ Filtering & Pagination
- ✓ Aggregations & Caching

Jobs API

Companies API

Skills API

Salaries API

Locations API

Auth API

Node.js • Express.js • SQL Server

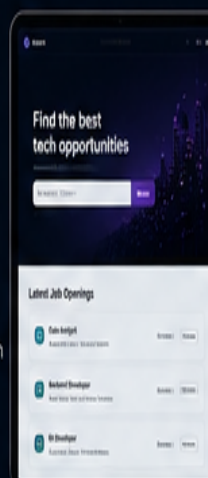
9/13



Frontend Web Application

A responsive web application built with React.js and Tailwind CSS.

- ✓ Job Listings
- ✓ Companies
- ✓ Skills
- ✓ Salaries
- ✓ Trends
- ✓ Dashboard
- ✓ User Authentication
- ✓ Favorites
- ✓ Comparisons



React.js • Tailwind CSS • Responsive UI

10/13



Admin Panel

Admin dashboard to manage users, applications and monitor the platform.



Dashboard Overview



User Management



Applications Management



Data Monitoring



Scrapers Logs



System Health



Email Templates



Monitor • Manage • Analyze

11/13



Power Automate (Registration Flow)

We automate user registration emails using Microsoft Power Automate.



- ✓ Trigger: When a new response is submitted
- ✓ Get response details
- ✓ Send welcome email to the user

12/13

End-to-End Architecture



Collect → Process → Analyze → Visualize → Automate

13/13

Tech Stack

Data & Analytics

- Python, Pandas
- SQL Server
- SSIS
- Power BI, DAX

Backend

- Node.js
- Express.js
- JWT
- REST API

Frontend

- React.js
- Tailwind CSS

Tools & Others

- Git & GitHub
- Postman
- VS Code
- Power Automate
- Microsoft Forms



TechPulse

Real-time Insights.
Better Decisions.
Smarter Careers.
Stronger Tech Community.

♥ Thank you!



TechPulse

EGYPT

Tech Job Market Intelligence Platform

Turning Tech Job Data into **Actionable Insights**

An end-to-end Data Engineering & Analytics platform — from web scraping and ETL pipelines to Power BI dashboards, a full-stack web app, and automated workflows.

Data Engineering

Analytics

Full-Stack

Automation

github.com/saraelshenawy632/TechPulse-Unlock-Your-Tech-Career

Open source on GitHub — explore the full codebase



Project Overview

What is TechPulse?

TechPulse is a data-driven platform that collects, analyzes and visualizes technology job market data to provide real-time insights and help people understand the tech job market better — bringing scattered, unstructured job data together in one intelligent platform.



Real-Time Job Data

Continuously collecting job postings from multiple platforms and sources.



Actionable Insights

Turning raw job data into useful, decision-ready intelligence.



Smarter Decisions

Helping job seekers and companies make better hiring & career choices.



Automated Registration

Instant welcome emails triggered the moment a user signs up.

Data-Driven

Real-Time

Automated



Data Collection & Cleaning

Gathering job data from multiple sources with Python



Python Web Scrapers

Custom scrapers built to collect job listings directly from company career pages and job boards.



APIs

REST API integrations for structured, reliable access to job market data.



Scheduled Jobs

Automated, recurring collection runs keep the dataset continuously up to date.



Data Cleaning & Transformation

Removing nulls, fixing errors, de-duplicating records and standardizing formats before storage.

Tools & Libraries

Python • Requests • BeautifulSoup • Selenium • Pandas

Data is gathered from multiple platforms and sources, then cleaned and validated before it ever reaches the warehouse.

Python

Automated Collection

Multi-Source



Data Engineering Architecture

SQL Server & the Data Warehouse

Data Flow

1. Raw Data → **2. Cleaned Data** → **3. Staging Area** → **4. Dimensional Data Warehouse**

Cleaned data is stored in SQL Server and structured into a dimensional model built for fast, reliable analytics.

Dimensional Model

Fact Table

FactJobs

Dimension Tables

DimCompany

DimLocation

DimPlatform

DimSkills

DimDate

FactJobSkills



Incremental ETL with SSIS

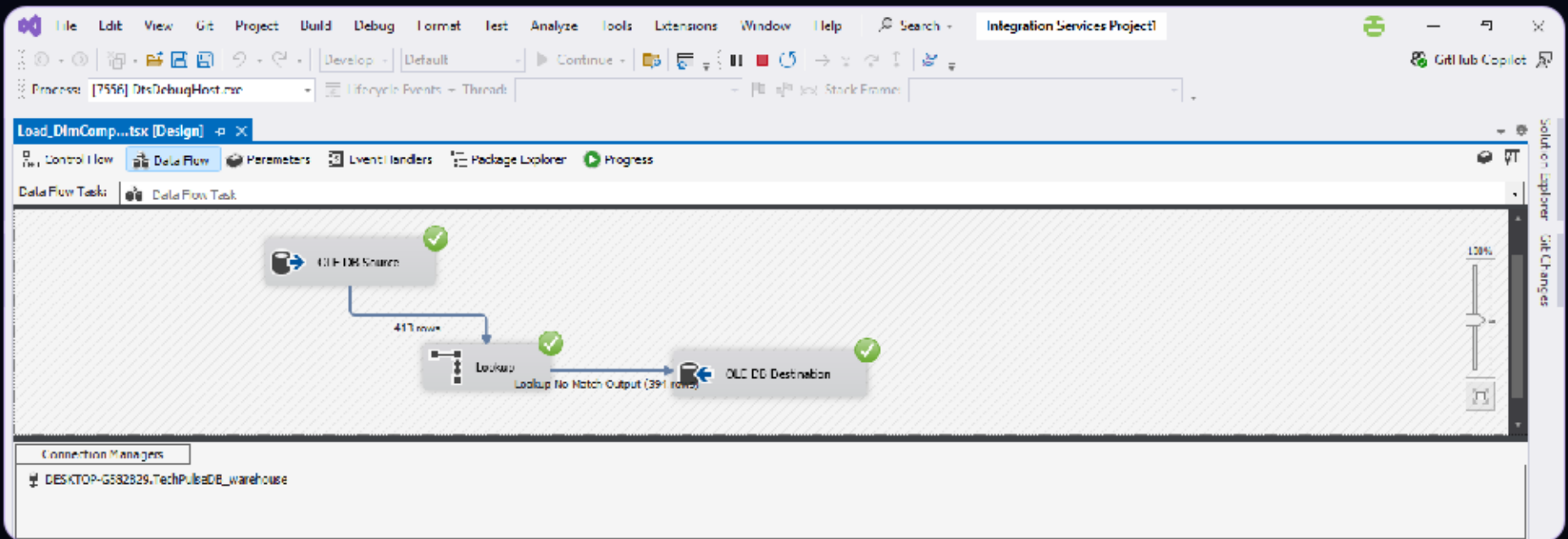
SQL Server Integration Services

Extract

Lookup

Identify New Records

Load



Live SSIS Package — Visual Studio (413 rows processed)



OLE DB Source

Extracts raw job data from the staging tables.



Lookup Transformation

Checks incoming rows against existing records.



OLE DB Destination

Writes validated, transformed rows into the warehouse.



Incremental Load

Only new or changed records are loaded — no duplicates.

Efficient

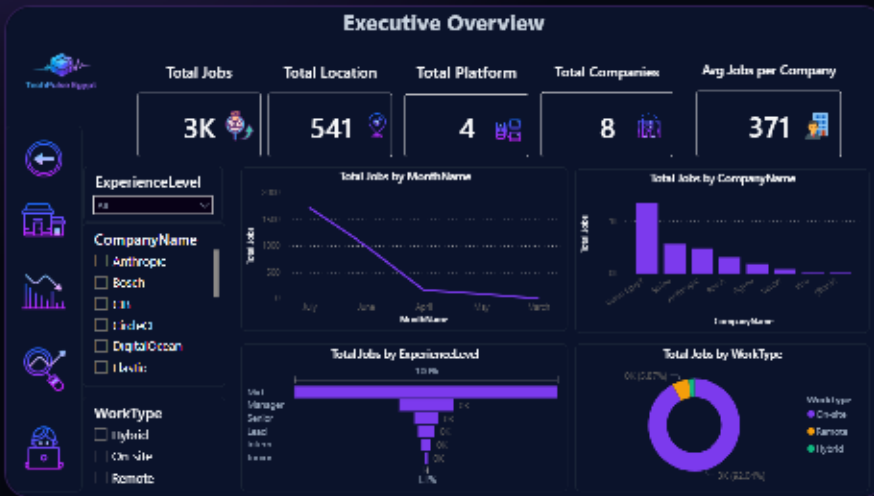
Reliable

Scalable

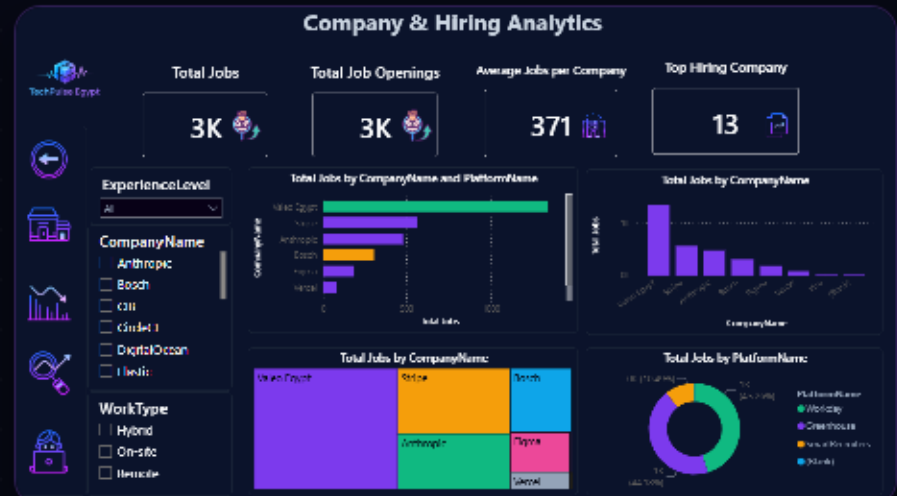


Power BI — Interactive Analytics

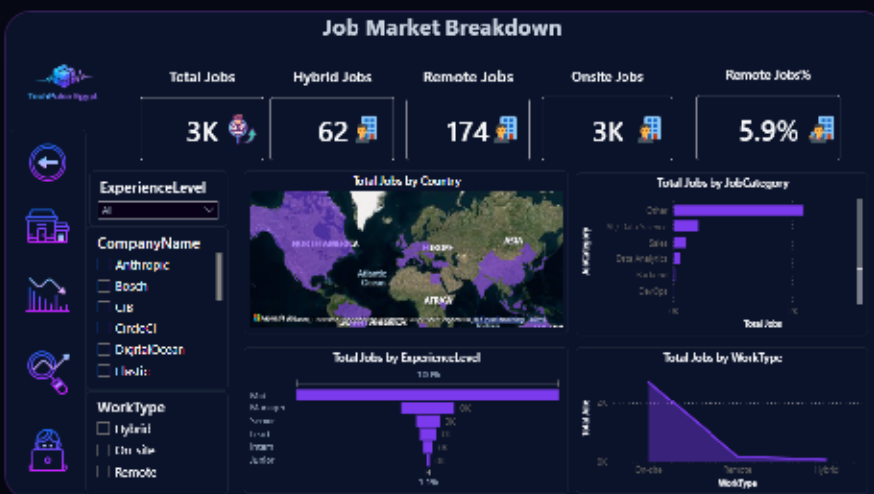
Turning warehouse data into decision-ready dashboards



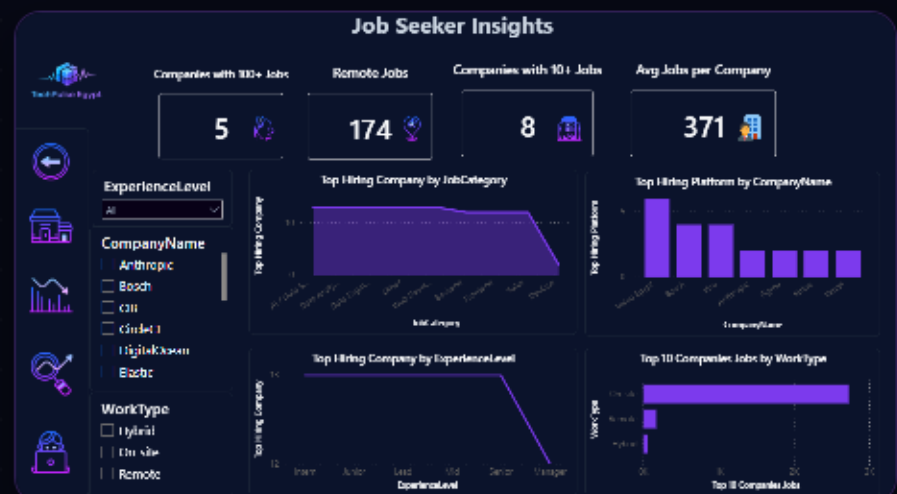
Executive Overview



Company & Hiring Analytics



Job Market Breakdown



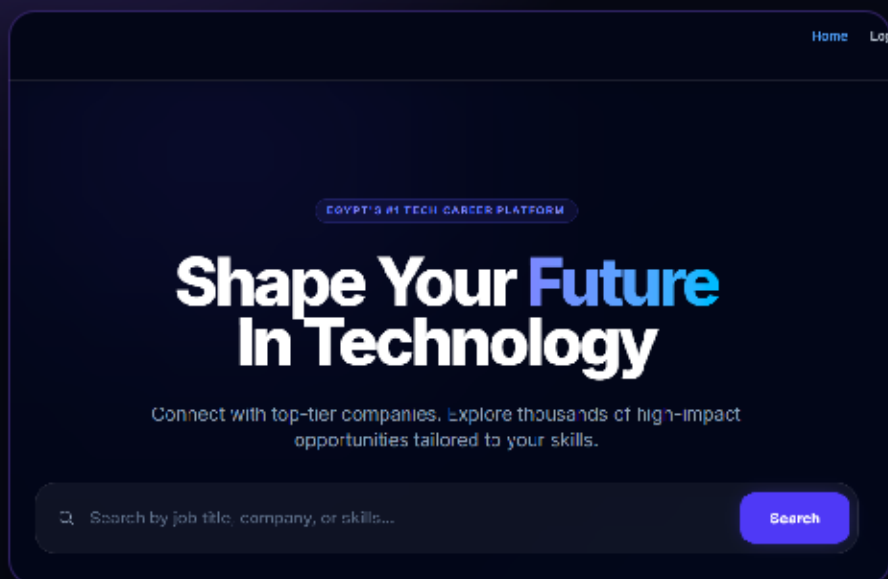
Job Seeker Insights

Four connected report pages cover hiring trends, company & platform analytics, remote vs. on-site breakdowns, and experience-level insights — all filterable in real time.

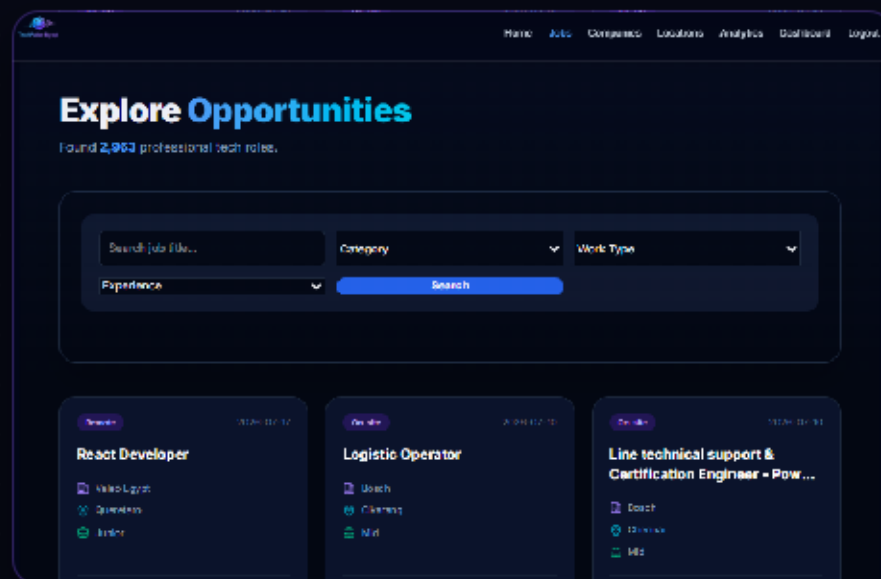


TechPulse Web Application

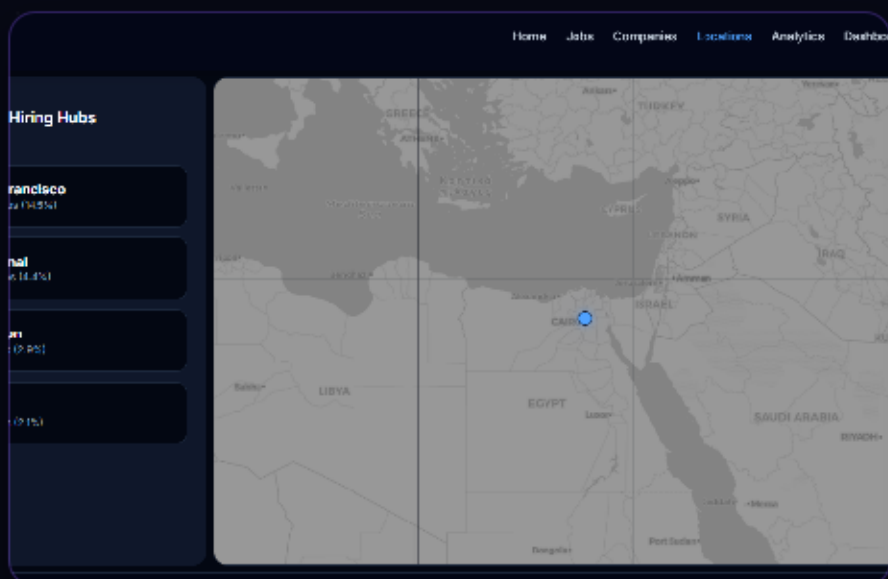
A responsive job platform built with React & Tailwind CSS



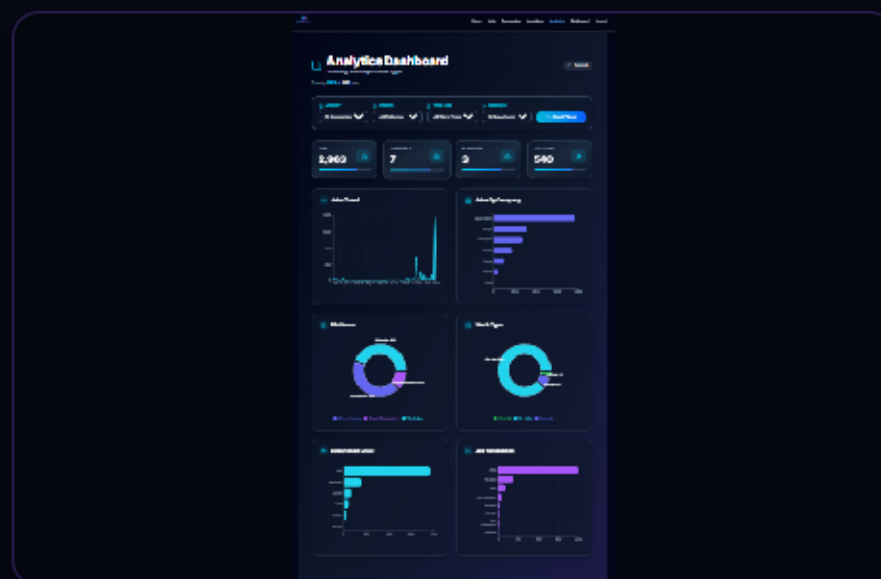
Landing Page



Explore Opportunities — Job Search



Locations — Hiring Hubs Map



Mobile — Fully Responsive



Smart Search & Live Map

Filter jobs by title, category, work type, experience & location.



Fully Responsive

The same rich experience adapts seamlessly from desktop to mobile.

React.js

Tailwind CSS

Responsive UI



Admin Panel

Monitor, manage and grow the platform



Dashboard Overview

The jobs management section shows a list of jobs with columns: JOB TITLE, CATEGORY, WORK TYPE, EXPERIENCE, POSTED, and ACTIONS. A '+ Add Job' button is visible. The table contains 7 rows of job listings.

JOB TITLE	CATEGORY	WORK TYPE	EXPERIENCE	POSTED	ACTIONS
Senior React Developer	Web Development	Hybrid	Senior	2026-07-17	Edit Delete
React Developer	Web Development	Remote	Junior	2026-07-17	Edit Delete
PO/PI Process/Manufacturing Engineering PTM	Other	On-site	Mid	2026-07-15	Edit Delete
Process/Manufacturing Automation Engineer	Other	On-site	Mid	2026-07-15	Edit Delete
Supply Chain Scheduler	AI / Data Science	On-site	Mid	2026-07-15	Edit Delete
Functional Safety Engineer - ADAS	Other	On-site	Mid	2026-07-15	Edit Delete
GTQ Junior Regulations & Norms Engineer	Other	On-site	Junior	2026-07-15	Edit Delete

Jobs Management

The applications tracking section shows a list of applications with columns: APPLICANT, JOB DETAILS, STATUS, and ACTIONS. A 'Total: 5 Requests' button is visible. The table contains 5 rows of application data.

APPLICANT	JOB DETAILS	STATUS	ACTIONS
sara alshenawy sara.alshenawy@gmail.com	Senior React Developer Web Development	Pending	View Accept Reject Cancel Delete
sara alshenawy sara.alshenawy@gmail.com	Fullstack Developer Web Development	Pending	View Accept Reject Cancel Delete
sara alshenawy sara.alshenawy@gmail.com	Fullstack Developer Web Development	Pending	View Accept Reject Cancel Delete
sara alshenawy sara.alshenawy@gmail.com	Use technical support & Certification Engineer - Power tools Web Development	Accepted	View Accept Reject Cancel Delete
sara alshenawy sara.alshenawy@gmail.com	Use technical support & Certification Engineer - Power tools Web Development	Pending	View Accept Reject Cancel Delete

Applications Tracking

The user management section shows a list of users with columns: NAME, EMAIL, ROLE, and ACTIONS. A search bar is visible. The table contains 4 rows of user data.

NAME	EMAIL	ROLE	ACTIONS
ahmed	ahmed@gmail.com	User	Edit Delete
ali	ali@gmail.com	User	Edit Delete
sara alshenawy	sara.alshenawy@gmail.com	Admin	Edit Delete

User Management



Monitor / Manage / Analyze

Track jobs, applications and users from a single real-time dashboard.

Monitor

Manage

Analyze



Application Review & Backend API

A robust REST API built with Node.js & Express.js

Home Jobs Companies Locations Analytics Dashboard Logout

[← Back](#)



sara elshenawy
selshenawy69@gmail.com

Job Title
Senior React Developer

Company
Valeo Egypt

Status
Pending

Application Actions

[✓ Accept](#)[✗ Reject](#)

Documents

[📄 View Resume](#)[🌐 LinkedIn Profile](#)

Timeline

🕒 Applied Date
7/18/2026, 5:59:32 AM

✅ Current Status
Pending

Reviewing an Application

API Highlights

[Jobs API](#)[Companies API](#)[Skills API](#)[Salaries API](#)[Locations API](#)[Auth API](#)

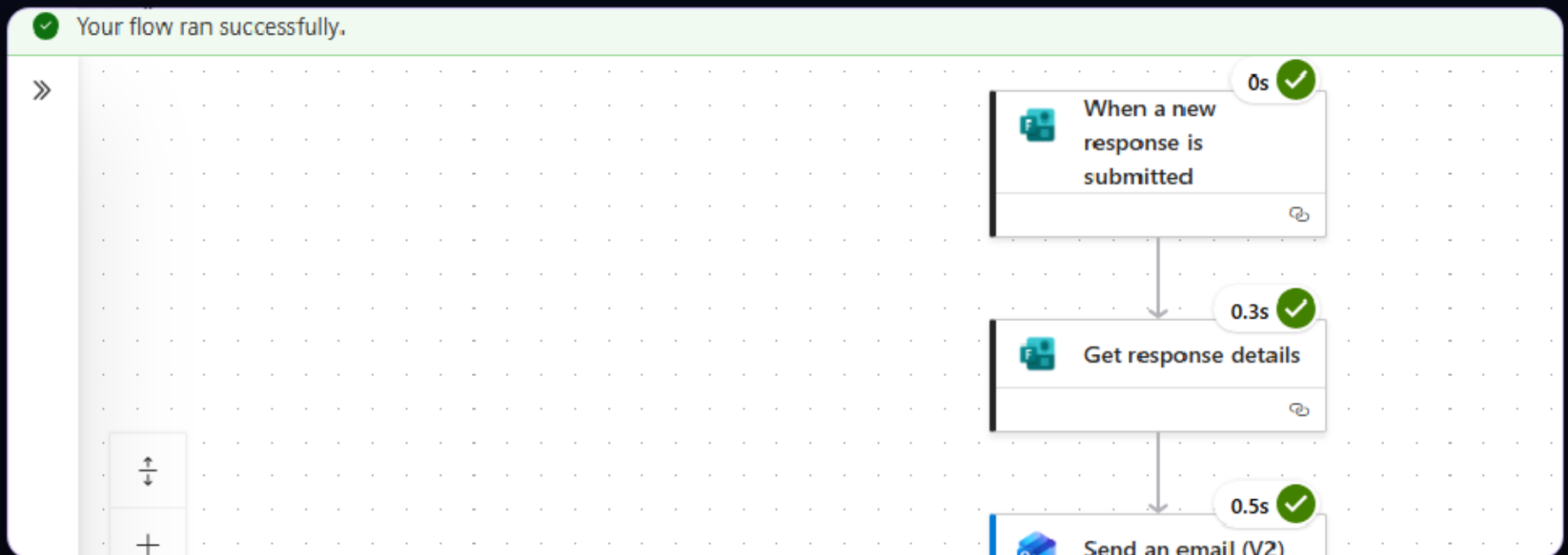
- ✓ JWT Authentication
- ✓ Rate Limiting
- ✓ Data Validation
- ✓ Filtering & Pagination
- ✓ Aggregations & Caching

[Node.js](#)[Express.js](#)[SQL Server](#)

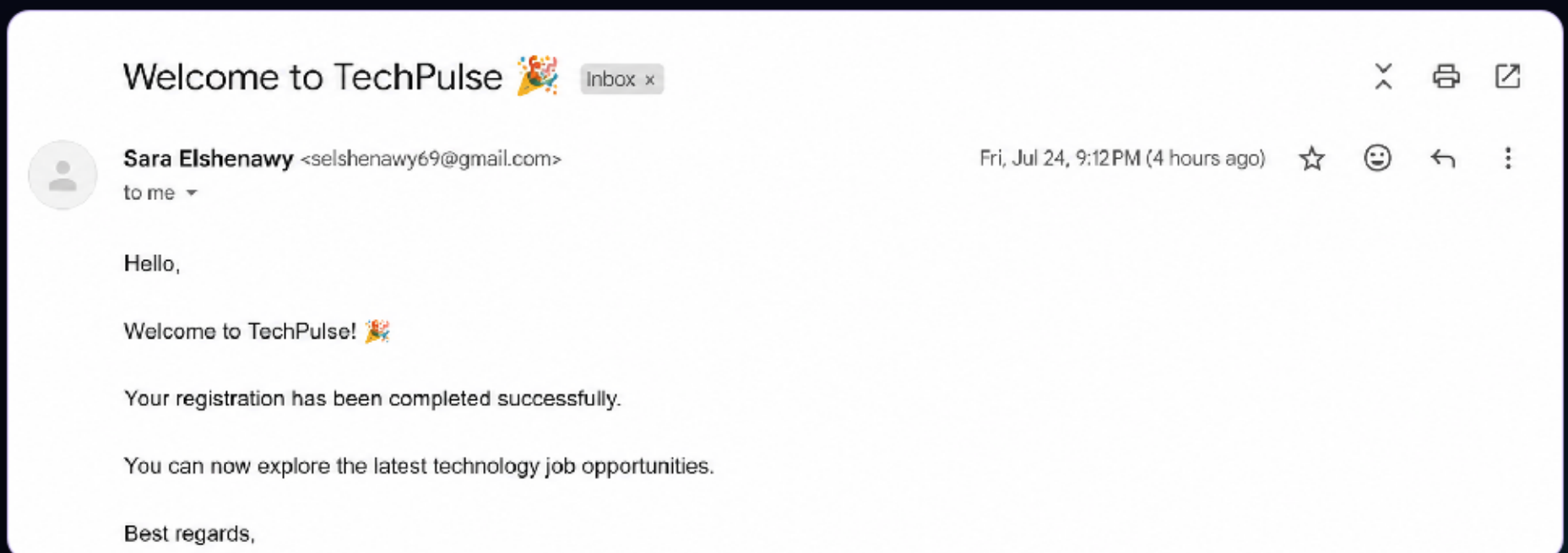


Registration Automation

Automating welcome emails with Microsoft Power Automate



Flow ran successfully



Welcome Email — Delivered

Power Automate

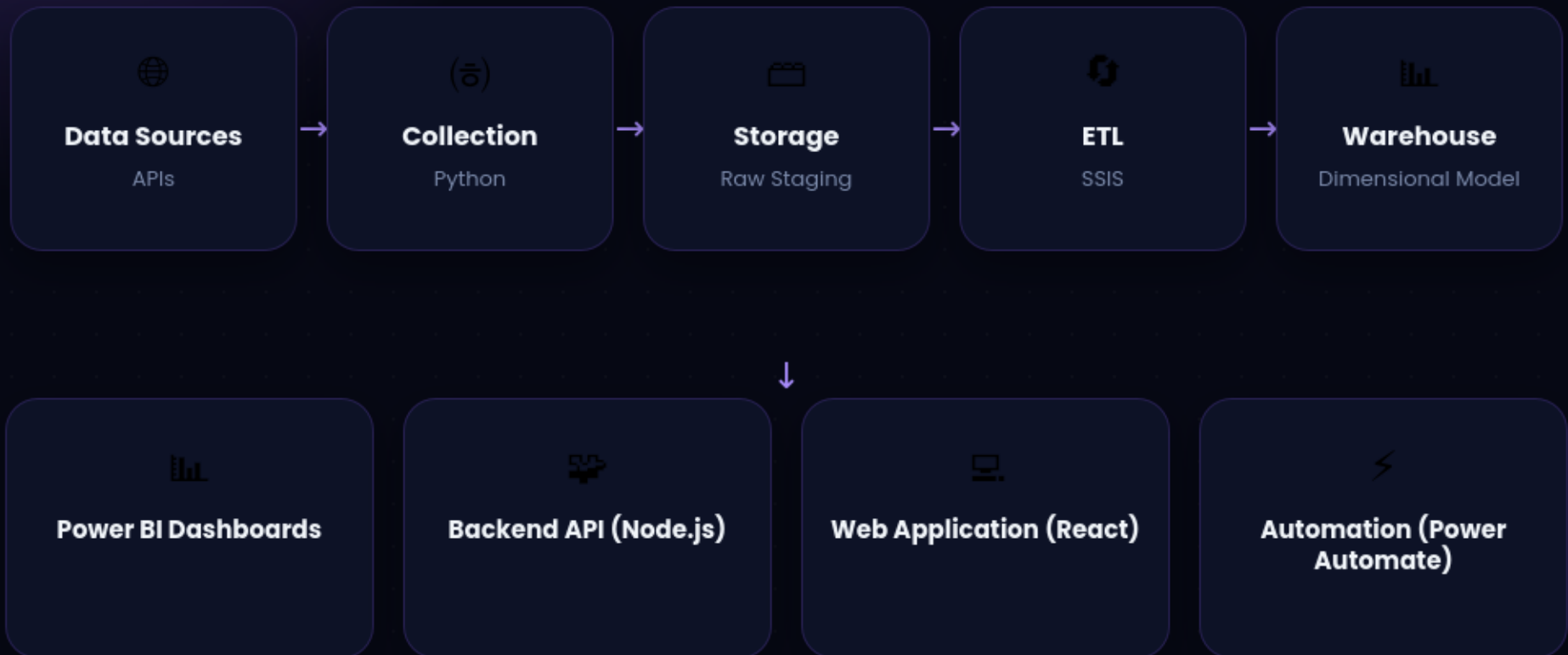
Zero-Touch Onboarding

Instant Delivery



End-to-End Architecture

From raw job postings to real-time insight



Collect → Process → Analyze → Visualize → Automate

Every layer of TechPulse works together: Python scrapers and APIs collect raw job data, SSIS incrementally loads it into a SQL Server dimensional warehouse, Power BI turns it into insight, and the full-stack web app & Power Automate deliver it — and act on it — in real time.



Tech Stack

The tools and technologies behind TechPulse

Data & Analytics

- Python, Pandas
- SQL Server
- SSIS
- Power BI, DAX

Backend

- Node.js
- Express.js
- JWT
- REST API

Frontend

- React.js
- Tailwind CSS

Tools & Others

[Git & GitHub](#)[Postman](#)[VS Code](#)[Power Automate](#)[Microsoft Forms](#)[Full-Stack](#)[Data Engineering](#)[Automation](#)



TechPulse

EGYPT

Tech Job Market Intelligence Platform

2,965

Jobs Tracked

8

Companies

4

Platforms

541

Locations

Real-Time Insights.
Better Decisions.
Smarter Careers.
Stronger Tech Community.

Built with Python • SQL Server • SSIS • Power BI • Power Automate • React • Node.js

github.com/saraelshenawy632/TechPulse-Unlock-Your-Tech-Career

♥ Thank you!